Accuracy as a Fraction

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

One connected accuracy record, with a labelled poll extension.

Score the baseline and candidate, change class mix, report missing coverage and attempt the new matrix independently.

Keep one habit: Keep cases, counts, positive class, denominators and limits linked.

MAIN LESSON · SECTION 1

Recall · Retrieve earlier evidence

> Keep the first attempt, relevant evidence and support used.

Your evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from **Training** and Testing. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Features and Distractors. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Debugging Strategies. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Training and Testing

Features and Distractors

Debugging Strategies

W1 · Score the training-majority baseline

> Retain its missed blocked cases.

Reference and your reasoning

- Main accuracy reference: training16 clear/4 blocked gives an always-clear baseline. On the same20 evaluation cases, reference rows and prediction columns are clear then blocked. Baseline [[16,0],[4,0]], candidate [[12,4],[0,4]]. Accuracy=(upper-left+lower-right)/all. Clear-correct fraction=upper-left/first-row-total; blocked-correct fraction=lower-right/second-row-total. Balanced accuracy averages those two fractions equally. Blocked positive: FP=upper-right, FN=lower-left. Undefined fractions are unavailable, not zero.

- **Training** has 16 clear and 4 blocked labels. State the majority rule. On the worked evaluation, give its matrix, **accuracy**, false positives and false negatives with blocked positive. Explain why 80 percent is incomplete.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Compare both class fractions

> Keep the two denominators visible.

Reference and your reasoning

- Main accuracy reference: training16 clear/4 blocked gives an always-clear baseline. On the same20 evaluation cases, reference rows and prediction columns are clear then blocked. Baseline [[16,0],[4,0]], candidate [[12,4],[0,4]]. Accuracy=(upper-left+lower-right)/all. Clear-correct fraction=upper-left/first-row-total; blocked-correct fraction=lower-right/second-row-total. Balanced accuracy averages those two fractions equally. Blocked positive: FP=upper-right, FN=lower-left. Undefined fractions are unavailable, not zero.

- For the worked candidate [[12,4],[0,4]], give ordinary **accuracy**, both **class-specific correct fraction**s, **balanced accuracy**, false positives and false negatives. Compare with the baseline.

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Change the class mix explicitly

> State the fixed arithmetic assumptions.

Reference and your reasoning

- Main accuracy reference: training16 clear/4 blocked gives an always-clear baseline. On the same20 evaluation cases, reference rows and prediction columns are clear then blocked. Baseline [[16,0],[4,0]], candidate [[12,4],[0,4]]. Accuracy=(upper-left+lower-right)/all. Clear-correct fraction=upper-left/first-row-total; blocked-correct fraction=lower-right/second-row-total. Balanced accuracy averages those two fractions equally. Blocked positive: FP=upper-right, FN=lower-left. Undefined fractions are unavailable, not zero.

- For the illustrative 8-clear 8-blocked set, use baseline [[8,0],[8,0]] and candidate [[6,2],[0,8]]. Calculate ordinary and **balanced accuracy** for each. Explain what stayed fixed and what this example does not prove.

> **My reasoning and evidence**

MAIN LESSON · SECTION 16

W5 · Report missing-class coverage

> Distinguish ordinary accuracy from an unavailable class fraction.

Reference and your reasoning

- Main accuracy reference: training16 clear/4 blocked gives an always-clear baseline. On the same20 evaluation cases, reference rows and prediction columns are clear then blocked. Baseline [[16,0],[4,0]], candidate [[12,4],[0,4]]. Accuracy=(upper-left+lower-right)/all. Clear-correct fraction=upper-left/first-row-total; blocked-correct fraction=lower-right/second-row-total. Balanced accuracy averages those two fractions equally. Blocked positive: FP=upper-right, FN=lower-left. Undefined fractions are unavailable, not zero.

- The supplied scores(counts) function requires nonzero counts for both reference classes and raises ValueError otherwise. It does not return ordinary accuracy in that branch; you may calculate ordinary accuracy separately from the observed counts. There is no measured blocked-class result when no blocked cases exist.

- An evaluation contains ten clear cases and no blocked cases. All **prediction**s are clear. State ordinary **accuracy**, explain why blocked performance is unavailable, and describe how our code handles the missing class.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

W4 · Assess the new comparison independently

> Keep the first attempt, relevant evidence and support used.

Your evidence record

- New **training** labels have a clear majority. A separate evaluation has 18 clear and 2 blocked records.

  The always-clear baseline is [[18,0],[2,0]] and candidate is [[15,3],[1,1]], with reference rows and clear then blocked on both axes. Calculate ordinary **accuracy**, both class fractions and **balanced accuracy** for each.

  Give candidate FP and FN, then explain why the metrics do not choose an automatic winner.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

My independent explanation, trace and limits

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Assess a new comparison independently

MAIN LESSON · SECTION 9

Project · Define the scored unit

> Keep the first attempt, relevant evidence and support used.

Your evidence record

- Report correct recommendations / all test users, alongside the raw test **count**.
- The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 22

Feedback · Repair and return

> Keep the first attempt, relevant evidence and support used.

Your evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Keep the first attempt, relevant evidence and support used.

Your evidence record

-
  - I can explain the mechanism.
  - I can show evidence.
  - I can handle a changed case.

- I can name a limit.

  Mark each as not yet, with help or independently, and attach supporting work.

- Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 response**

REFERENCE · SECTION 2

Reference · Supported W2 response

> Use when helpful; keep practice distinct from the reserved first attempt.

Additional evidence

Use W2: Compare both class fractions for **Accuracy** and **class imbalance**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair a score claim

REFERENCE · SECTION 3

Extra practice · Repair a score claim

> Use when helpful; keep practice distinct from the reserved first attempt.

Additional evidence

Use the worked case for **Accuracy** and **class imbalance**. Proposed explanation: “High **accuracy** means both classes work well”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original target

REFERENCE · SECTION 5

Reference · Original target

> Use when helpful; keep practice distinct from the reserved first attempt.

Additional evidence

Class A gets 8/10 cases right and class B gets 18/30 right. Compute each rate and the combined rate. Why is averaging the two **percentage**s misleading here?

▼ **Extra practice · Simple accuracy case**

REFERENCE · SECTION 6

Extra practice · Simple accuracy case

> Use when helpful; keep practice distinct from the reserved first attempt.

Additional evidence

Nine of twelve held-out classifications are correct.

Compute **accuracy** and state the **denominator**.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ **Extra practice · Changed score case**

REFERENCE · SECTION 7

Extra practice · Changed score case

> Use when helpful; keep practice distinct from the reserved first attempt.

Additional evidence

A classifier gets 9 of 12 cases right. Another gets 8 of 10 right. Calculate both accuracies and identify the larger fraction.

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 12 Test and reflect

▼ **Extra practice · A revealing comparison**

REFERENCE · SECTION 8

Extra practice · A revealing comparison

> Use when helpful; keep practice distinct from the reserved first attempt.

Additional evidence

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ Extra practice · Earlier transfer

REFERENCE · SECTION 11

Extra practice · Earlier transfer

> Use when helpful; keep practice distinct from the reserved first attempt.

Additional evidence

Class A gets 8/10 cases right and class B gets 18/30 right. Compute each rate and the combined rate. Why is averaging the two **percentage**s misleading here?

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Same selected W4 response copy

REFERENCE · SECTION 15

Reference · Same selected W4 response copy

> Use when helpful; keep practice distinct from the reserved first attempt.

Additional evidence

New **training** labels have a clear majority. A separate evaluation has 18 clear and 2 blocked records.

The always-clear baseline is [[18,0],[2,0]] and candidate is [[15,3],[1,1]], with reference rows and clear then blocked on both axes. Calculate ordinary **accuracy**, both class fractions and **balanced accuracy** for each.

Give candidate FP and FN, then explain why the metrics do not choose an automatic winner.

My reasoning and evidence

▼ Extension W1 · State the known denominator

REFERENCE · SECTION 17

Extension W1 · State the known denominator

> Use after the main lesson or in a separately scheduled denominator session.

Reference and your extension record

- Poll extension reference: worked responses are Yes, No, Yes, Unknown, Yes. Yes among known means Yes/(Yes+No); disclose Unknown separately. Yes among all means Yes/all supplied. If there are no known responses, report no known responses rather than0%. W4 uses its own supplied sequence. This extension has its own assessment and feedback; it is not the main classification W4.

- In the worked poll, why is the known **denominator** 4 rather than 5?

> **My reasoning and evidence**

▼ **Extension W2 · Compare proportions and counts**

REFERENCE · SECTION 18

Extension W2 · Compare proportions and counts

> Use after the main lesson or in a separately scheduled denominator session.

Reference and your extension record

- Poll extension reference: worked responses are Yes, No, Yes, Unknown, Yes. Yes among known means Yes/(Yes+No); disclose Unknown separately. Yes among all means Yes/all supplied. If there are no known responses, report no known responses rather than0%. W4 uses its own supplied sequence. This extension has its own assessment and feedback; it is not the main classification W4.

- Poll A has 2 Yes of 4 known and Poll B has 3 Yes of 6 known. Compare **proportion**s and **count**s.

> **My reasoning and evidence**

▼ **Extension W3 · Handle no known responses**

REFERENCE · SECTION 19

Extension W3 · Handle no known responses

> Use after the main lesson or in a separately scheduled denominator session.

Reference and your extension record

- Poll extension reference: worked responses are Yes, No, Yes, Unknown, Yes. Yes among known means Yes/(Yes+No); disclose Unknown separately. Yes among all means Yes/all supplied. If there are no known responses, report no known responses rather than0%. W4 uses its own supplied sequence. This extension has its own assessment and feedback; it is not the main classification W4.

- All three responses are Unknown. What should the program report?

My reasoning and evidence

▼ **Extension W4 · Summarize the new poll independently**

REFERENCE · SECTION 20

Extension W4 · Summarize the new poll independently

> Use after the main lesson or in a separately scheduled denominator session.

Reference and your extension record

- Poll extension reference: worked responses are Yes, No, Yes, Unknown, Yes. Yes among known means Yes/(Yes+No); disclose Unknown separately. Yes among all means Yes/all supplied. If there are no known responses, report no known responses rather than0%. W4 uses its own supplied sequence. This extension has its own assessment and feedback; it is not the main classification W4.

- **Process** Yes, Yes, No, Unknown, No, Yes. Report Yes **count**, known **count**, unknown **count** and Yes **proportion** among known.

My reasoning and evidence

▼ Extension W5 · Limit a volunteer-poll claim

REFERENCE · SECTION 21

Extension W5 · Limit a volunteer-poll claim

> Use after the main lesson or in a separately scheduled denominator session.

Reference and your extension record

- Poll extension reference: worked responses are Yes, No, Yes, Unknown, Yes. Yes among known means Yes/(Yes+No); disclose Unknown separately. Yes among all means Yes/all supplied. If there are no known responses, report no known responses rather than0%. W4 uses its own supplied sequence. This extension has its own assessment and feedback; it is not the main classification W4.

- Three of five known volunteers said Yes. Can you say 60 percent of all students agree?

My reasoning and evidence

▼ Extra practice · Shelf-repair picture

REFERENCE · SECTION 23

Extra practice · Shelf-repair picture

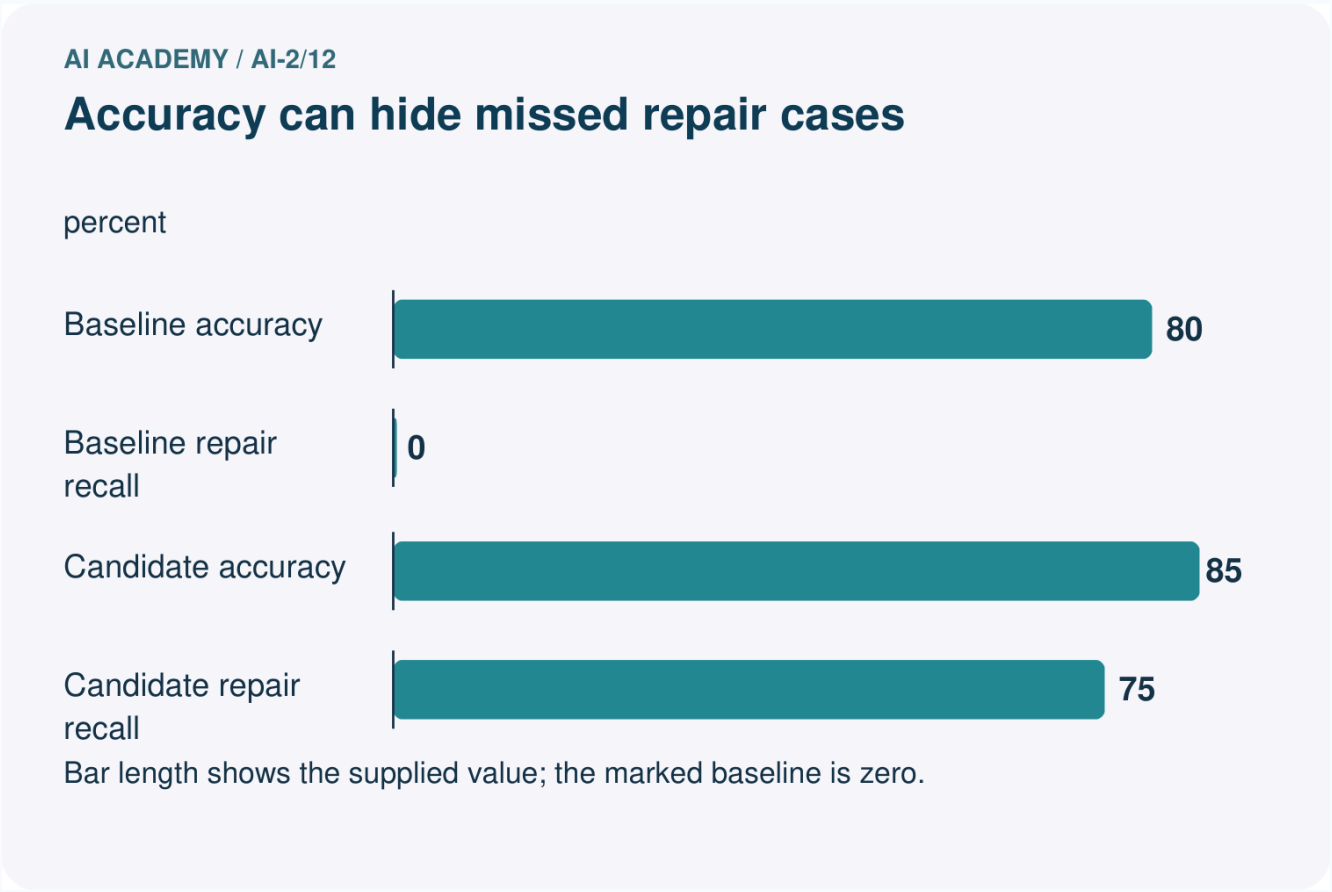
> Use when helpful; keep practice distinct from the reserved first attempt.

Additional evidence

A fictional shelf classifier predicts repair needed or not needed.

Evaluation reference labels: 16 not-needed and 4 needed. Positive class is needed.

Always-not-needed baseline is correct on 16. Candidate has TP=3,FN=1,FP=2,TN=14.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/12

Accuracy as a Fraction

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Reference population

20 labelled cases, including four repair-needed books.



2. Baseline

16/20=80% accuracy; catches 0/4 needed cases.



3. Candidate

(3+14)/20=85% accuracy; catches 3/4=75% needed cases.



4. Compare costs

Candidate also raises two false alarms; discuss both missed repairs and extra checks.

CHANGE ONE THING

A candidate instead marks every book repair needed.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why is 100% repair detection not sufficient to declare the changed classifier best?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Compare recommender choices with a simple baseline and report relevant rare-category failures, not only aggregate **accuracy**.

▼ **Reference · Portfolio milestone**

REFERENCE · SECTION 24

Reference · Portfolio milestone

> Use when helpful; keep practice distinct from the reserved first attempt.

Additional evidence

Project: Transparent Book Recommender

Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Compare recommender choices with a simple baseline and report relevant rare-category failures, not only aggregate **accuracy**.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Pictured evidence and repair

REFERENCE · SECTION 25

Extra practice · Pictured evidence and repair

> Try before opening feedback; keep independent W4 separate.

First question

- A fictional shelf classifier predicts repair needed or not needed.
- Which labelled step matches this detail? Candidate also raises two false alarms; discuss both missed repairs and extra checks.
- 1. Reference population
- 2. Compare costs
- 3. Candidate
- 4. Baseline
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after trying**

After your attempt, compare with the supported explanation: Compare costs Candidate also raises two false alarms; discuss both missed repairs and extra checks.

A majority-class baseline can look accurate while missing every positive case. The candidate improves overall agreement and repair detection in this table, but false alarms still matter.

Use the same cases, **denominator**s and positive-class definition for both. Do not infer that a single metric identifies the best policy for every library.

Changed-case question

- Changed-case practice: A candidate instead marks every book repair needed. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. Recall is 4/4=100%, but **accuracy** is only 4/20=20%.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the change**

After your attempt, compare with the supported explanation: Recall is 4/4=100%, but **accuracy** is only 4/20=20%. All 16 not-needed books become false alarms.

Repair and keep your work

Review **Training** and Testing (ai-2/11). Goal: Choose a split unit that matches the intended **prediction** claim. Short practice: How many distinct books are in each repaired role, and why report that as well as scan **count**s?.

Return to this week and explain what you changed.

Review Features and Distractors (ai-2/9). Goal: Choose recorded properties that preserve distinctions needed for a stated task. Short practice: Does this one pair prove the classifier always ignores subject?

Explain a narrower claim.. Return to this week and explain what you changed.

Review Debugging Strategies (ai-2/6). Goal: Design revealing tests for a function. Short practice: Which test most directly isolates the handling of loose books, and what should it produce?.

Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

▼ Reference · Six-picture response sheets

REFERENCE · SECTION 26

Reference · Six-picture response sheets

> Choose the picture serving the current question.

01. Accuracy as a Fraction: the case

AI ACADEMY / AI-2/12 · 01 · LOOK AT THE FACTS

Accuracy as a Fraction: the case

The situation

A fictional shelf classifier predicts repair needed or not needed.

Supplied evidence

Evaluation reference labels: 16 not-needed and 4 needed. Positive class is needed. Always-not-needed baseline is correct on 16. Candidate has TP=3,FN=1,FP=2,TN=14.

By the end, I can compare accuracy with class-specific performance in an imbalanced evaluation.

Follow the labelled picture with the matching explanation. Open the vector diagram

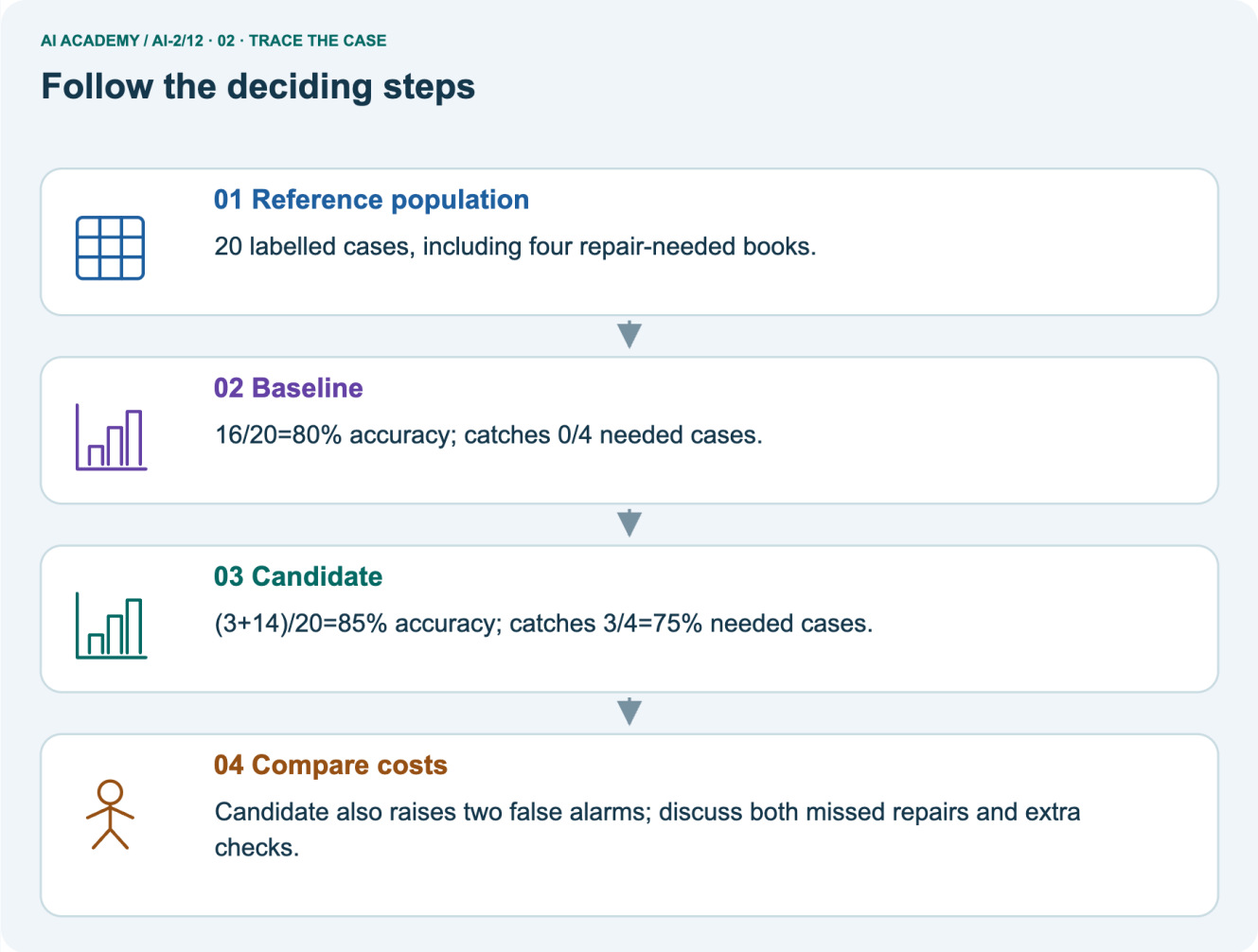
A fictional shelf classifier predicts repair needed or not needed. Supplied facts: Evaluation reference labels: 16 not-needed and 4 needed. Positive class is needed. Always-not-needed baseline is correct on 16.

Candidate has TP=3,FN=1,FP=2,TN=14.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Reference population: 20 labelled cases, including four repair-needed books. → Baseline: 16/20=**80% accuracy**; catches 0/4 needed cases. → Candidate: (3+14)/20=85% **accuracy**; catches 3/4=75% needed cases. → Compare costs: Candidate also raises two false alarms; discuss both missed repairs and extra checks.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Accuracy can hide missed repair cases



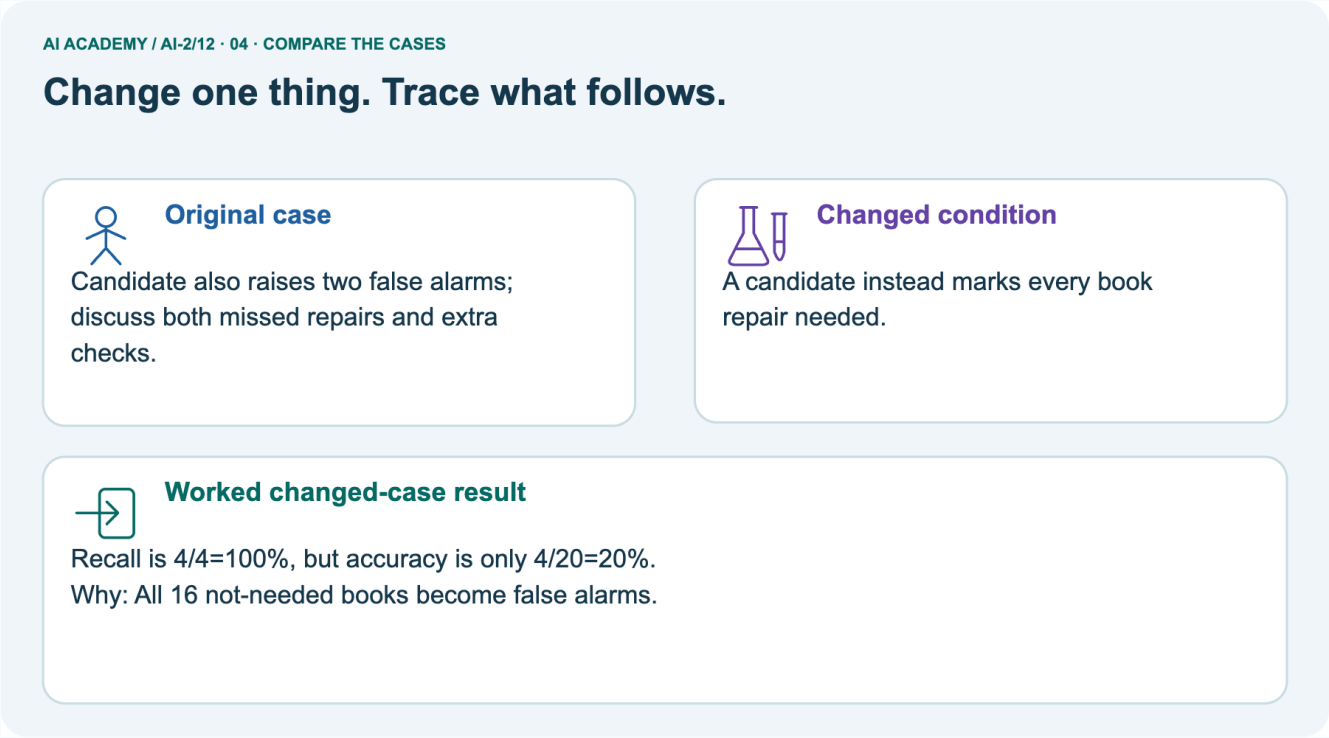
Follow the labelled picture with the matching explanation. Open the vector diagram

Accuracy can hide missed repair cases. Units: percent. Baseline **accuracy**: 80. Baseline repair recall: 0. Candidate **accuracy**: 85. Candidate repair recall: 75.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

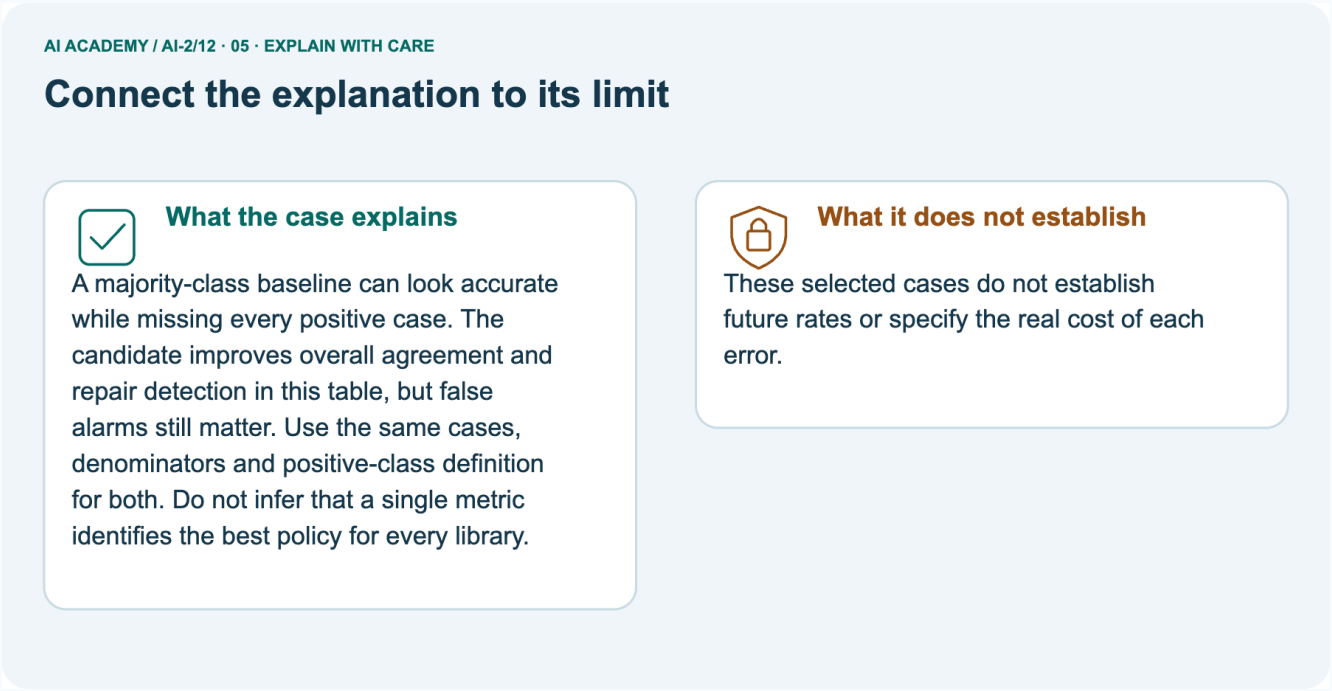
Original case: Candidate also raises two false alarms; discuss both missed repairs and extra checks. Changed condition: A candidate instead marks every book repair needed.

Expected result: Recall is 4/4=100%, but **accuracy** is only 4/20=20%. Reason: All 16 not-needed books become false alarms.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A majority-class baseline can look accurate while missing every positive case. The candidate improves overall agreement and repair detection in this table, but false alarms still matter.

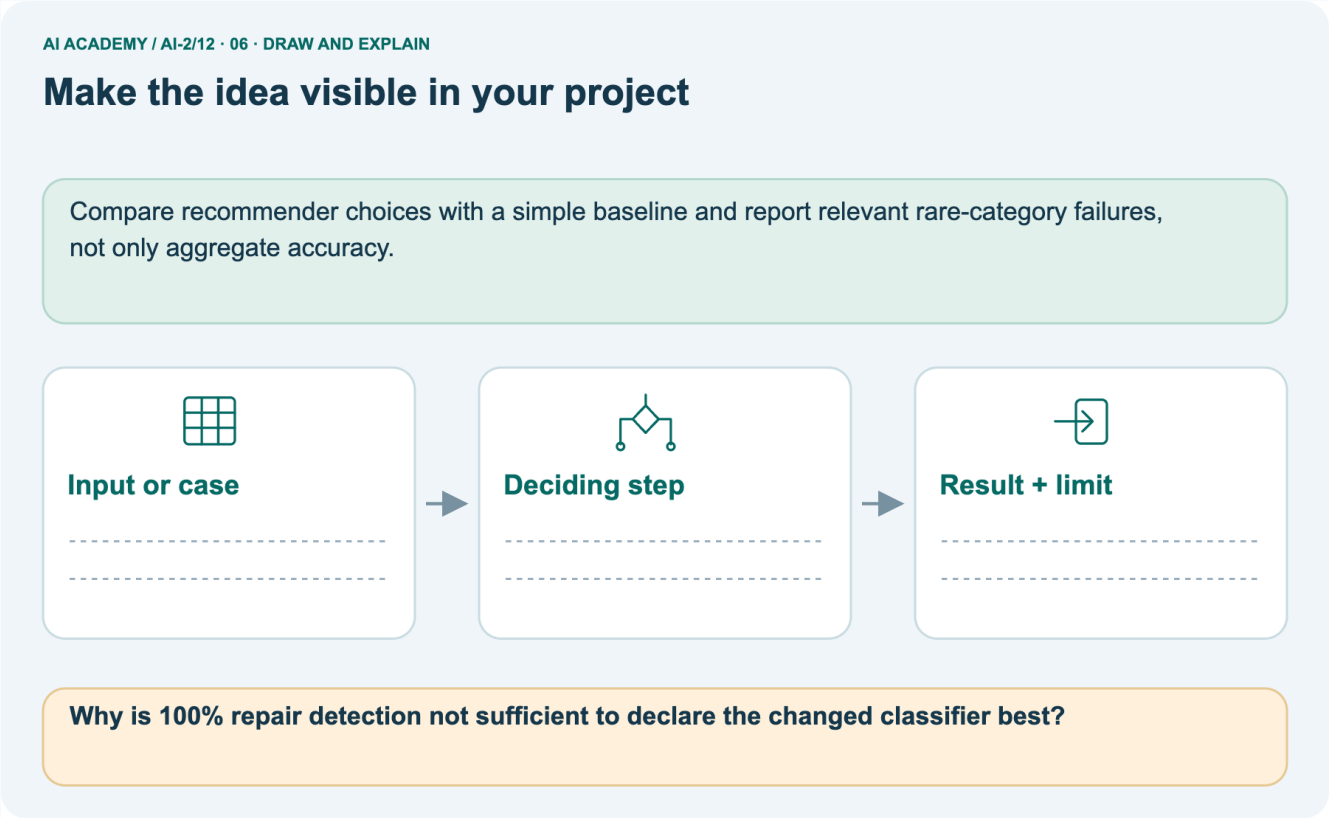
Use the same cases, **denominator**s and positive-class definition for both. Do not infer that a single metric identifies the best policy for every library.

Limit: These selected cases do not establish future rates or specify the real cost of each error.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Compare recommender choices with a simple baseline and report relevant rare-category failures, not only aggregate **accuracy**. Draw your **input**, deciding step, result and limit.

Supported question: Why is 100% repair detection not sufficient to declare the changed classifier best?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Accuracy as a Fraction: the case

02. Follow the deciding steps

03. Accuracy can hide missed repair cases

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.